

Session Objectives

1. Understand the epidemiology, pathophysiology, and clinical presentation of **cystic fibrosis**, **asthma**, **pneumonia** and **bronchiolitis**
2. Distinguish between multiple causes of respiratory distress in newborns, **including transient tachypnea of the newborn** and **meconium aspiration syndrome**
3. Identify radiographic differences between common pediatric respiratory pathologies
4. Distinguish between multiple causes of pediatric upper airway disease, **including epiglottitis**, **retropharyngeal abscesses**, and **peritonsillar abscesses**
5. Understand the social factors in pediatric pulmonology, including environmental health and vaccine non-adherence

Source: Course Syllabus

Case 1

A 15-month-old toddler, presents to your general pediatrics office for the first time with a wet cough for a “long time”.

His mother reports a history of recurrent pneumonia, one episode of which required hospitalization.

She also reports that he seems small for his age. He sometimes has bowel movements that smell really bad.

The patient was adopted. Birth history: unremarkable, biological parents: unknown

He lives with his adoptive parents in a home built in 2002.

Clinical Signs and Symptoms

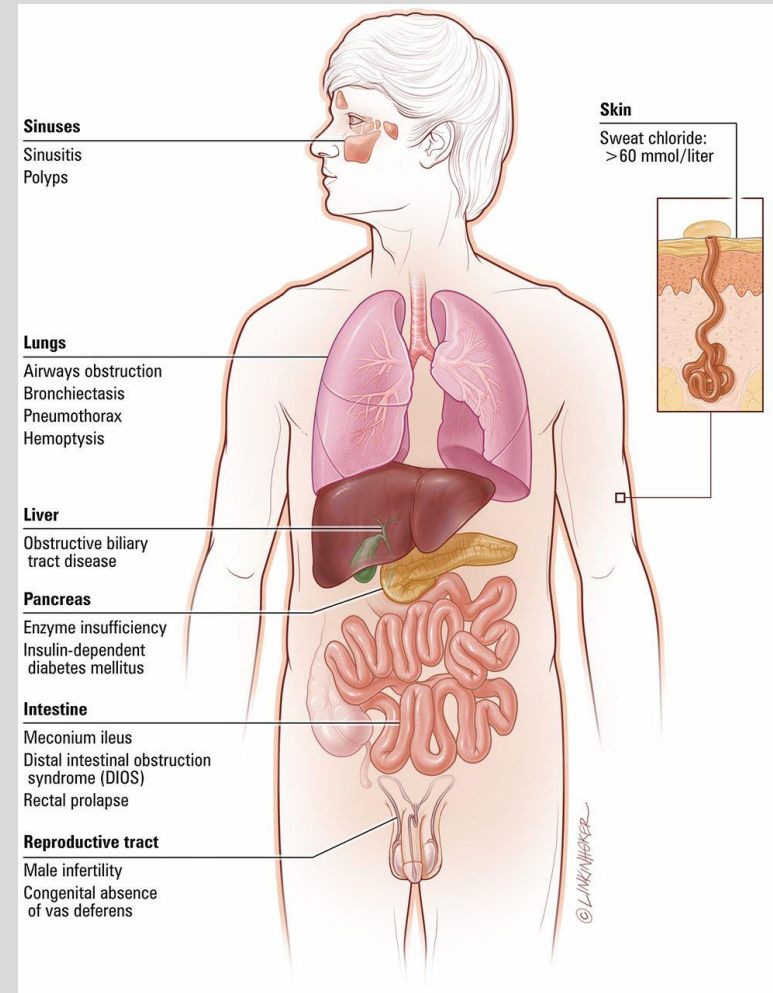
- Neonatal/GI:
protein/sugars(less so)
and esp. fat
malabsorption, failure to
thrive, steatorrhea, liver
disease
- Pulmonary: chronic
cough, chronic resp.
infection, nasal polyps,
recurrent sinusitis,
pneumonia
- Other:
 - Vitamin deficiency (A,
D, E, K-fat soluble)
 - CF-related diabetes
 - Infertility
 - Electrolyte
disturbances
 - Digital clubbing
 - Congenital absence of
vas deferens

Case 1: Diagnosis

- Newborn screening: all 50 states
- **Sweat Cl testing(positive: >60meq or mmol/L)**
- Criteria:
 - ≥ 1 typical clinical feature AND
 - 2 elevated sweat Cl tests OR
 - 2 known CF mutations OR
 - abnormal nasal epithelial potential difference

Case 1: Some Indications for Sweat Cl testing

- Respiratory:
 - Chronic/recurrent cough
 - Recurrent bronchiolitis
 - Recurrent wheeze/difficult to control asthma
 - Unexplained respiratory *P. aeruginosa*
- Gastrointestinal:
 - Meconium ileus
 - Neonatal intestinal obstruction
 - Steatorrhea, malabsorption
- Miscellaneous:
 - Positive newborn screen
 - Digital clubbing
 - Failure to thrive
 - Family history



Text adapted from Nelson's Essentials of Pediatrics Table 137.2, 2023

Image: Pediatrics in Review May 2014, VOLUME 35 / ISSUE 5

Case 1: Management

- Respiratory: clearing airway secretions
 - Chest physiotherapy
 - OMM
 - Aerosolized DNase + saline (nebulized)
 - Infection: aggressive IV antibiotics
- Pancreatic insufficiency:
 - Enzyme capsules (lipase, protease)
 - Vitamin supplementation (A, D, E, K)
- Meconium Ileus: enemas
- CF-related diabetes: insulin
- Cholestasis: ursodeoxycholic acid
- Even well-controlled patients have a long problem list
- 2019 CFTR gene modulator approved by the FDA. For some has improved overall symptoms.

Case 2

A 9 yo child presents to the Emergency Department with his parent.

He complains of a 10-hour history of wheezing and shortness of breath.

His mother states he had “a bad cold” for the past 3-4 days.

He was given one albuterol nebulizer treatment at home an hour ago with no relief.

Case 2: Physical Exam

General: Ill-appearing, in distress, seated upright with hands on knees and leaning forward

Vitals: Temp 99F, HR 124, BP 128/94, RR 38, Pulse Oximetry: 90% on room air

Resp: sternocleidomastoids appear hypertonic bilaterally, wheeze appreciated on exhalation in all lung fields, and slight upper-lobe inspiratory crackles

Cardio: +S1/ S2 heard, no MRG

Asthma: Epidemiology

- 4.7 million children(CDC) under 18(6.2% '21) in the U.S.
- Children: males > females (opposite in adulthood), approx. 8% overall population. Stable numbers but disparities exist.
- Past 20 years: **2-7x** more ED visits and hospitalizations (2017 1.6 mil ED and 183,000 hospitalizations). increased deaths in black children. ED/UCC visits higher in South and large metro areas.
- 2016-2018 CDC: death rates higher among adults, females and blacks- Northeast, Midwest and West. Good news for the South and young people. Higher in rural vs urban.
- Associations: small and medium metro areas, low income, Black, Puerto Rican, allergic conditions
- Strongest risk factors: atopy, family history

Return to the Patient

Pt is administered a nebulized formulation of albuterol + ipratropium bromide. Three hours after initial presentation, repeat vitals:

T: 98.8F, HR: 90, RR: 20, SpO2: 98% (room air)

When you ask a more detailed history, pt. and his mom report that he has shortness of breath each day at recess or gym class, wakes up two nights a week wheezing, and “doesn’t always get to the end of the month” with each inhaler.

How would you classify our patient’s asthma?

Case 2: DX/Management

- Diagnostic: spirometry(>5yr of age)
- Stepwise therapy, multiple drug classes
 - Inhaled corticosteroids (↓morbidity, may affect growth)
 - Leukotriene modifiers ([1] Leukotriene receptor antagonists, [2] leukotriene synthesis inhibitors)
 - Long acting beta-2 agonists (Formoterol, Salmeterol)
 - Theophylline, cromolyn sodium (rarely used)
 - Biologics, monoclonal antibodies (anti-IgE, anti-IL5, anti-IL5R, anti-IL4/IL13)
 - Short-acting beta-2 agonists (albuterol)
 - Anticholinergics (Ipratropium)
 - Oral(systemic) corticosteroids (avoid prolonged use)

Case 3

A female born at 38wks gestation from a primigravid mom via SVD is noted to have darkly stained and foul-smelling amniotic fluid upon delivery.

You note low tone and cyanosis of the extremities. Apgar scores are 5 and 6 at 1 and 5 minutes, respectively.

Neonatal resuscitation is performed, and the infant is admitted to the NICU.

Case 3: What is the diagnosis?

- A. Meconium Aspiration Syndrome
- B. Transient Tachypnea of the Newborn
- C. Pneumonia
- D. Hydrops Fetalis
- E. Neonatal Respiratory Distress Syndrome

Meconium Aspiration Syndrome (MAS)

- Seen in 5% of newborns with meconium-stained amniotic fluid
- Aspiration → airway obstruction → respiratory distress
- X-ray: infiltrates, streaking, ↑A-P diameter, flattened diaphragms
- Meconium effects on the lungs:
 - Ball-valve effect
 - Chemical irritation
 - Predisposition for infection
 - Inhibition of production and function of surfactant
- Often seen with primary pulmonary hypertension of the newborn (↑pulmonary vascular resistance)

Differential Diagnosis: Newborn Respiratory Distress

- Preterm
 - Respiratory Distress Syndrome (“Hyaline Membrane Disease”) (RDS)
 - Apnea of prematurity
- Full-term and post-term
 - Primary Pulmonary Hypertension of the Neonate (PPHN)
 - Meconium Aspiration Pneumonia
- Preterm and full-term
 - Transient Tachypnea of the Newborn
 - Bacterial Sepsis (Group B *Streptococcus*)

Adapted from Table 61.1 in Nelson’s Essentials of Pediatrics (pathologies denoted as common)

Respiratory Distress: Lab Evaluation

- Chest X-Ray:
 - Parenchymal pattern determination, ?pneumothorax, cardiac morphology
- Arterial Blood Gas:
 - Acid-base disturbance, extent of hypercapnia/hypoxia
- Complete Blood Count:
 - H+H, white count, platelet count (?DIC)
- Blood Culture:
 - Recovery of offending microbe
- Blood Glucose:
 - Hypoglycemia, stress hyperglycemia
- Echocardiogram:
 - Interrogation of murmur or PPHN, cardiomegaly, structural heart disease

Adapted from Table 61.2 in Nelson's Essentials of Pediatrics

Case 4

Vitals: HR 166, RR 48, Temp 100.2F, O2 Sat. 90% on room air

General: in mild distress with noticeable increased work of breathing

HEENT: clear nasal d/c with alar flaring, rest is normal

CV: Reg. rhythm, S1 and S2 present without S3, S4, MRG

Resp: supraclavicular retractions, diffuse wheeze and crackles heard throughout all lung fields

Bronchiolitis: Etiology + Epidemiology

- Causes:
 - Respiratory syncytial virus (**RSV**)
 - Others: human metapneumovirus, parainfluenza, influenza, adenovirus, rhinovirus, coronavirus
- Usually ≤ 2 years of age (peak: 2-6 mos old)
- Increased risk: premature infants, chronic lung disease, congenital heart disease, neuromuscular weakness, immunodeficiency
- FDA 7/17/23: Beyfortus(nirsevimab) monoclonal antibody against RSV, IM single dose 0-24 mos old

Bronchiolitis: Natural History + Clinical Signs

- Course:
 - Incubation (4-6 days)
 - 3-7day progression to noisy breathing, wheezing, increased work of breathing (WOB)
 - Pulmonary dysfunction and bronchial hyperreactivity may persist for several years
- Signs: prolonged expiration, wheeze, nasal flaring, suprasternal retractions, air trapping + lung hyper-expansion
- Grunting and cyanosis = severe disease
- Young infants can present with apnea

Pneumonia: Management

- Symptoms, physical exam, imaging, age will typically direct the care
- Etiology isolated/identified 40-80% of the time
 - More urgent to discern in hospitalized patients
- Empyema or pleural effusion: thoracentesis is diagnostic and therapeutic
- Surgical management: abscesses, pneumatocele, empyema
- **For this patient, pneumonia can trigger a new asthma exacerbation, and asthma can predispose a pneumonia**

Case 6: PE

General: hoarse cry, inspiratory stridor, suprasternal retractions, toxic appearing

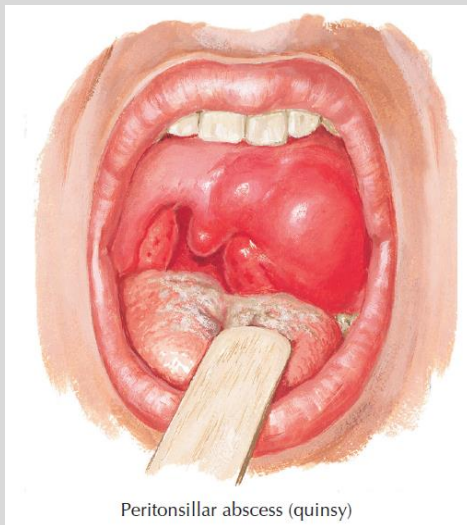
Vitals: T 102 F, HR 150 bpm, RR 40 bpm, O2 sat 97% (room air)

Resp: High pitched noise over trachea on inspiration, lungs CTA bilaterally without wheezes, rales, or rhonchi

CV: normal S1 and S2 with no S3, S4, murmurs, rubs, or gallops

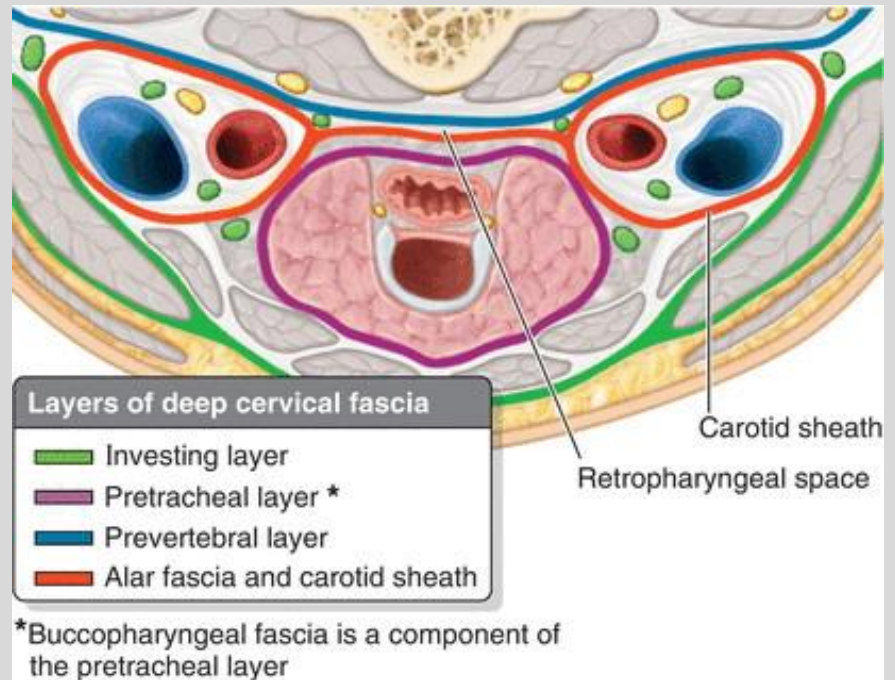
Other causes of Stridor

- Retropharyngeal abscess
- Foreign body aspiration
- Laryngotracheobronchitis (croup)
- Peritonsillar abscess
- Airway hemangioma
- Subglottic stenosis
- Vocal cord paralysis



Peritonsillar abscess (quinsy)

Netter's Pediatrics Fig. 35-4



Moore's Clinically Oriented Anatomy 8e Fig. 9.6

Summary Slide

- **Cystic Fibrosis:** multiple mutations → sweat Cl testing, multiorgan involvement, chronic condition, improving lifespan
- **Asthma:** reversible, episodic; may need *maintenance* therapy; many triggers
- **Meconium Aspiration Syndrome:** multiple effects on lungs; different from other newborn resp. distress causes
- **Bronchiolitis:** usually RSV, peak 2-6 mos. of age
- **Pneumonia:** age group helps predict etiology, guides treatment; viral vs. bacterial clinical and x-ray patterns
- **Epiglottitis:** often vaccine-preventable, inflammation of epiglottis on lat. Neck x-ray, clinical signs and PE differentiate from other causes of respiratory distress